THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 01/10/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

High assembly error rates and operational bottlenecks caused by factory technicians relying on static paper SOP manuals during assembly.

2. AI RATIONALE & ANALYSIS

"Augmented Reality replaces static paper SOPs with real-time, hands-free visual overlays directly onto the physical workpiece to eliminate procedural errors. THEXRA View provides rapid deployment of structured spatial work instructions with enterprise-grade reliability tailored for industrial factory operations."

3. PROPOSED SOLUTION CONCEPT

Deploy a hands-free AR visual work instruction solution that projects contextual 3D step-by-step guidance and component placement cues directly onto factory assembly parts.

IMPLEMENTATION APPROACH:

1. Phase 1: SOP Digitization - Convert critical paper-based assembly manuals into standardized spatial AR work instructions using THEXRA Make.
2. Phase 2: Workstation Pilot - Deploy wearable AR headsets at two pilot assembly stations to validate ergonomic usability and step accuracy.
3. Phase 3: Production Rollout - Integrate with plant MES infrastructure and scale across remaining production lines with frontline training.

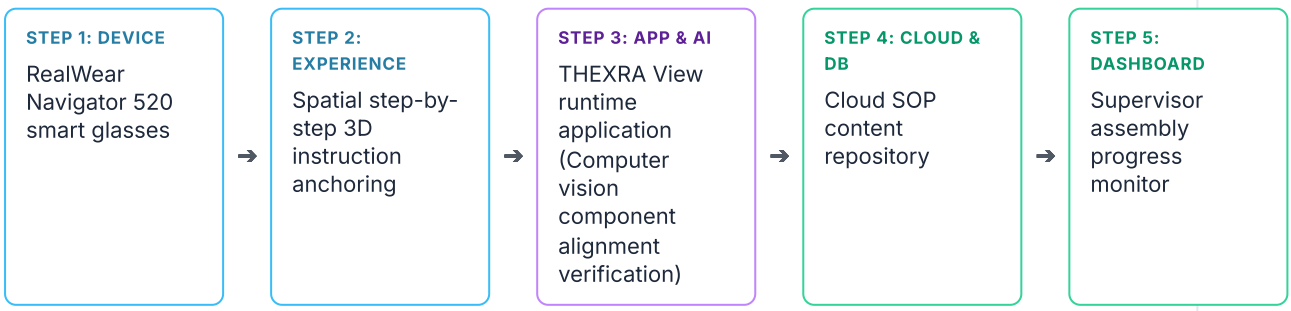
DETAILED IMPLEMENTATION PLAN:

1. **Discovery** (1 week):Factory assembly line audit report • Technical requirements and network readiness document
2. **Design** (1.5 weeks):Spatial SOP UX wireframes • Digital instruction template standards
3. **Prototype** (1.5 weeks):Pilot AR assembly instruction module • Technician usability feedback report
4. **Development** (3 weeks):Complete digital SOP library build • Centralized content management integration
5. **Testing** (1.5 weeks):On-floor acceptance testing results • Hardware battery and ergonomic validation report
6. **Deployment** (1.5 weeks):Factory floor production rollout • Operator and supervisor training sign-off
7. **Support** (Ongoing):Continuous platform maintenance SLA • Monthly defect reduction and usage report

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

AR — THEXRA View

AUTOMATED SYSTEM FLOW DIAGRAM:



> **1. Overview:** A wearable AR assembly guidance system delivering real-time, interactive procedural instructions to frontline operators. The platform connects directly to centralized work instruction repositories to ensure factory-wide SOP version consistency.
>
> **2. User Journey:** Technician logs into AR headset and scans the workstation barcode → Visual 3D step-by-step overlays appear directly over physical parts → Technician verifies each step using voice or gesture commands → Assembly logs automatically sync with factory quality systems upon completion.
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> **3. Hardware:** RealWear Navigator 520 smart glasses, Ruggedized industrial tablets for backup access, Stationary barcode and QR scanners
>
> **4. Software:** THEXRA View runtime application, THEXRA Make procedural authoring tool, Device management client
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> **5. AI Components:** Computer vision component alignment verification, Voice command recognition module
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> **6. XR Components:** Spatial step-by-step 3D instruction anchoring, CAD-based part highlighting and animated overlays, Head-tracking and gaze-based navigation
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> **7. Backend:** Cloud SOP content repository, REST API gateway for MES integration, Role-based access control and security service
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> **8. Dashboard:** Supervisor assembly progress monitor, Error frequency and step-cycle-time analytics, Centralized SOP version control management
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> **9. Data Flow:** AR Wearable Device → Local Edge Gateway → Central SOP Backend & MES → Supervisor Management Dashboard
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> **Alternative Option:** Combination solution — THEXRA Suite (Remote, Make, View)

5. EXPECTED BUSINESS OUTCOMES

Up to 40% reduction in assembly defects, 30% reduction in technician training time, and complete digital traceability for assembly quality compliance.

6. IMPLEMENTATION NEXT STEPS

Conduct an on-site workflow audit of existing paper SOPs and select two high-complexity assembly stations for a scoped pilot implementation.